Personality Words or Life Circumstances?

Probing Human-Like Rule-Bending in LLM Respondents with a 2016 Japanese Survey of Everyday Rights Violations

Ryo Tanohata
Independent researcher
[TODO: email]

Draft, 31 October 2026

Abstract

Large language models (LLMs) are increasingly used as stand-in survey respondents. We ask whether such respondents show the situational "rule-bending" that a 2017 master's thesis identified as a core of human nature: people step outside rules in gray zones after weighing their own position and circumstances. We had Claude models answer the thesis's 54-item questionnaire on everyday rights violations (copyright, ownership, privacy; 5-point scale, Japanese) in pre-registered survey experiments (720 simulated respondents, about 32 million tokens), and tested the main result inside an LLM-agent village in a third (about 66 million tokens). The only human reference available is the thesis's published principal-component table ($n = 84$), which is internally inconsistent; we compare against the share implied by its first eigenvalue (0.268). *Experiment 1* ($n = 290$): without a persona, respondents were near-identical (mean item SD 0.14–0.37; H1a supported) and far from the thesis's "common-sense" answers (H2 not supported). Big Five personality words increased spread (H3 supported) but collapsed it onto a single strictness axis (first-component share 0.392, CI excluding 0.268; strictness correlated +0.78 with conscientiousness), and only one of the five human components re-appeared (H5 not supported). Asking for a candid first impression (*honne*) before each answer made answers *stricter*, the opposite of the prediction (H4). *Experiment 2* ($n = 230$) replaced personality words with first-person life circumstances (age, job, money, family, media habits). The first-component share fell to 0.228 (CI containing 0.268; H6 and H10 supported), students and financially strained respondents judged downloading more leniently (H9 supported), and Opus respondents also diversified (H11 supported). Of the thesis's four conditions that encourage rule-bending, "everyone around me does it" and "I have never heard of anyone being caught" made gray-zone answers more lenient, whereas "I am in a weak position" and "it is technically easy" did not (H8 partially supported). The human component structure was still not reproduced (H7 not supported). A follow-up that changed the topic of the surroundings to borrowing moved the borrowing answers but not the first component, which in all life-circumstance conditions was nearly the same strictness axis. An exploratory, dictionary-based coding of 1,150 one-phrase reasons found the thesis's rule-bending rationales in 31–57% of reasons, with "follow the majority" rising sharply under "everyone does it". *Experiment 3* gave the household representatives of a simulated village the same kind of information—their household's circumstances, and "what others around here have been doing"—in three copies of the village run for three simulated years; none of six pre-registered hypotheses was supported, and rule-related choices (keeping a private store, sharing, proposing penalties or taxes, judging disputes) were essentially unchanged. We conclude that personality words produce a one-dimensional caricature, whereas concrete life circumstances produce a more human-like spread and partly human-like situational adjustment on paper—but not the human content of that adjustment, and not, in our village, a change in behaviour.

1 Introduction

"We must humanize technology before it dehumanizes us." The author's master's thesis [13] opened with this sentence, attributed there to the neurologist Oliver Sacks, as its motivation.[1] The thesis asked what, concretely, the "humanity" is that technology might take away, and answered with a model and a survey.

The model combined Maslow's theory of motivation [9] with Freud's structural theory [5]. Humans have unconscious basic needs (physiological, safety, belonging and love, esteem, self-actualisation, and the need to know); this layer—the *id*—contains both good and bad impulses. The needs become socially acceptable behaviour only after censorship by the *superego* and adjustment by the *ego*; which form the satisfaction takes is shaped by environment, culture and custom. The survey (84 Japanese adults, 54 items on everyday rights violations) and interviews (13 participants) found five kinds of awareness (rights, ownership, information, privacy, self-responsibility) and, crucially, that people do not apply rules mechanically. In gray zones they adjust situationally, using seven ways of thinking: *if there is a legitimate reason, one may use it freely*; *take responsibility for one's own actions*; *act if the damage is tolerable*; *an act without malice is forgivable*; *follow the majority*; *where conditions are clear, take them into account and act so that no one is harmed*; *where conditions are unclear, interpret them in one's own favour*. Rule-bending was more likely when "everyone around me does it", when one "has never heard of anyone being caught", when one is "in a weak position", and when it is "technically easy"; it was less likely when creators would lose income, when a clear system existed, and out of respect for creators. The thesis concluded that law decides by rules, while people decide as subjects who weigh conditions—and that this situational adjustment is the human nature at stake.

Today the question can be turned around. LLMs are routinely asked to role-play human respondents [1, 2, 6]. If they reproduce the *average* answer but not the situational adjustment, they are exactly the kind of technology that flattens the humanity the thesis tried to describe. A companion observational study of a century-long LLM-agent village [14] found agents that almost never chose harmful actions, converged on forgiveness, and rarely gave themselves excuses to bend rules. This paper tests that impression in a controlled survey setting against the thesis's published results.

We ask: (i) do LLM respondents vary like a population, and in the same *shape* as the human sample? (ii) Do personality descriptions (Big Five) or concrete life circumstances produce more human-like variation? (iii) Do the thesis's four "rule-bending conditions" shift LLM answers in the human direction? (iv) Which of the thesis's seven ways of thinking appear in LLM respondents' own reasons? (v) Does information that shifts answers on paper also shift choices made by LLM agents inside a simulated society?

Contributions.

- Two pre-registered experiments and a pre-registered follow-up (720 simulated respondents, three Claude models) on a Japanese questionnaire about everyday rights violations, with all decision thresholds fixed before data collection and all hypotheses reported, including non-supported and opposite-direction results.
- Evidence that Big Five personality words produce variation that is dominated by one "strictness"

[1] We could not locate this sentence in Sacks's published writings or interviews by web search (29 October 2026); a near-identical phrase is widely associated with the affective-computing researcher Rana el Kaliouby in talks from about 2018 onwards. We therefore cite it as quoted in the thesis and do not attribute it further. [TODO: author: give the source the thesis used, or keep this footnote.]

axis, whereas first-person life circumstances without any character words bring the first-component share into the human range.

- A factorial test of four human rule-bending conditions: two shift LLM answers in the human direction, two do not.
- A transparent, LLM-free dictionary coding of 1,150 one-phrase reasons against the thesis's seven ways of thinking (exploratory).
- A pre-registered transfer test inside an LLM-agent village (three arms, one copy each, three simulated years): the circumstances that moved survey answers did not move rule-related choices (a null result).

2 Related work

LLMs as simulated respondents. Argyle et al. [2] conditioned GPT-3 on demographic backstories of real survey respondents and argued for "algorithmic fidelity": such "silicon samples" reproduced fine-grained subgroup response patterns. Horton [6] proposed treating LLMs as *homo silicus* to pilot behavioural-economics experiments, and Aher et al. [1] introduced "Turing experiments" that replicate classic human-subject studies with simulated participants. Dillion et al. [4] discussed when language models might replace human participants in psychology. Mei et al. [10] found chatbot behaviour in economic games often within the human range but shifted towards the altruistic and cooperative. Critical work has documented limits: Santurkar et al. [11] showed substantial misalignment between LM opinions and US demographic groups, even when steered; Bisbee et al. [3] found that synthetic survey responses had too little variance and were sensitive to prompt wording and model version; Wang et al. [15] argued that identity-prompted LLMs misportray and *flatten* identity groups. Our "too uniform" and "one-axis" findings are in line with this literature, but our reference is a non-US, non-political, everyday-ethics survey.

Personas and personality in LLMs. Serapio-García et al. [12] administered validated personality inventories to LLMs and showed that Big Five levels in generated text can be shaped by prompting. Jiang et al. [8] gave GPT models Big Five personas and found that self-reported scores followed the assigned profiles with large effects. Hu and Collier [7] quantified the "persona effect" in subjective annotation tasks and found that persona variables explain little of the human variance and that persona prompting yields only modest gains. Our Experiment 1 shows a related failure mode: persona words do steer answers, but along a single global axis rather than the item-specific structure seen in humans.

Companion study. This paper is the second of two. The first [14] is an observational account of a 108-simulated-year LLM-agent society; its "humanness" observations (forgiveness lock-in, absence of violence, model-dependent behaviour) motivated the present, pre-registered survey probe. Experiment 3 uses copies of that village.

3 Methods

3.1 Questionnaire and human reference

The questionnaire is the thesis's "Survey on property and other rights of others" (2016): 35 items on copyright, information and devices (A01–A35) and 19 on ownership and everyday life (B01–B19).

Each describes a behaviour (e.g., "Downloading music from a commercial CD for free from an individual's website") to be rated on five options: *a serious problem* (5), *somewhat a problem* (4), *cannot say* (3), *not much of a problem* (2), *no problem at all* (1). Appendix Table 4 lists all items with English glosses. The thesis also gave, for each item, an expected "common-sense" answer; we use these as a *norm* (not as human data). We call 14 items *gray-zone* items: those with expected answers "cannot say" or "not much of a problem", plus the five items probed in the thesis's interviews (A08, A09, A23, B08, B11).

Human reference: published table only. The raw human answers are not available to us. We use only the thesis's published principal-component table (Table 3; 84 valid respondents; PCA on standardized items): eigenvalues 14.47, 4.51, 3.18, 2.82, 2.20, and the top five items (with signs) of each of five components, named rights, ownership, information, privacy and self-responsibility awareness. **The table is internally inconsistent**: for 54 standardized items the eigenvalues sum to 54, so the first component explains $14.47/54 = 0.268$ and five components 0.503, whereas the printed cumulative proportions are 0.36 and 0.81. We could not resolve the discrepancy and use the eigenvalue-based values (0.268) as the reference, reporting the printed ones alongside. Because item means and SDs of the human sample are unavailable, several comparisons (H1a, H2) are against fixed thresholds rather than against humans.

3.2 Respondents and conditions

Each simulated respondent is one independent call to a Claude model through a Claude Code workflow agent, which reads one questionnaire file and returns a JSON object constrained by a schema (all 54 items required, integers 1–5). Respondents share nothing. Items are always presented in the original order. Models were addressed by the aliases `sonnet`, `haiku` and `opus`; the exact model versions behind the aliases on 29 October 2026 were not recorded [TODO: author: record versions if retrievable]. Sampling parameters (e.g., temperature) were the platform defaults and could not be set.

Experiment 1 (pre-registered at commit `2c8f966c`). Five conditions (Table 1): no persona ("answer with your own views") with Sonnet, Haiku and Opus (C0, C3, C4; 30 each); Big Five personas with Sonnet (C1, 100); and the same 100 personas asked to write a one-line candid first impression (*honne*: "what would you want to do and why") before each numeric answer (C2, 100; paired with C1). Personas had age, sex, prefecture and five independently drawn Big Five z-scores rendered as verbal levels ("very low" $<$ 10th percentile, "low", "average", "high", "very high") with a one-line description (e.g., high conscientiousness: "plans ahead, keeps rules and promises"). Trait intercorrelations were set to zero because we could not verify published values. All 290 respondents answered; none were re-asked. Cost: about 12.8 million tokens (plus 0.44 million for a 10-respondent pilot that is excluded from analysis).

Experiment 2 (pre-registered at commit `b88ad84c`). Personality and moral-character words were banned (checked automatically). Instead each respondent received a first-person description of their life (8–11 sentences) generated from assumed distributions: age, sex, prefecture, position (student, regular or non-regular employee, self-employed, homemaker, not working), financial situation (tight / ordinary / comfortable), household, digital literacy, how they usually listen to music, whether they have close friends with whom "even if something happens, we can talk it over later and understand each other" (53%), and three thesis-derived conditions that *discourage* rule-bending: a creator in the family or among friends (17%), buying works to support creators

(28%), and having been taught the rules at work or school (49%). An example (translated): "I am a 29-year-old woman living in Fukuoka. I work as a regular employee in IT system development and live with my parents. I am comfortable and can buy books and music without worrying. ... I usually listen to music on a monthly streaming service. I have close friends and colleagues nearby with whom I often lend and borrow things ... I have a favourite singer and buy their new songs to support them."

D1 (Sonnet, 100) received only the description; D2 (Sonnet, 100; same descriptions, paired with D1) additionally received "recent surroundings": four lines, one per thesis rule-bending condition, each stated either on the bending side or on the opposite side (both sides written out, so that "extra text" is constant). The 16 combinations were balanced (each condition on the bending side for 50 respondents; maximum inter-condition correlation 0.04). The bending-side lines were, in translation: "Among my friends and colleagues it has become normal to get music and videos for free and casually post each other's photos and writing"; "I have never heard of anyone around me being caught or warned for copying or unauthorised use online"; "Lately I often feel in a weak position: an unexpected expense left me short of money, and I find it hard to speak up"; "With a smartphone or PC, saving music or video, sharing photos or copying others' text now takes one button, and I know how". D3 (Opus, 30; descriptions S001–S030) tested whether the most uniform model of Experiment 1 also diversified. After all 54 numeric answers, every Experiment 2 respondent wrote a one-phrase reason (about 20 characters) for the five interview items. All 230 respondents answered. Cost: about 10.1 million tokens (plus 0.26 million for a 6-respondent pilot, excluded).

Experiment 3: inside the village (pre-registered at commit `b1c1ae8a`). The companion village [14] is a rule-based world (food, stores, births and deaths, neighbouring settlements) in which household representatives, played by Claude models, answer at seasonal and yearly meetings in a fixed JSON format. We copied the village state at simulated day 12,599 (year 104) three times and ran each copy for three simulated years (8 meetings, 12 representatives each, 96 answers per arm). World rules, random seed and answer format were identical; only the representatives' prompt differed. **A0** (control) used the unchanged prompt. **A1** added a section "your household's circumstances" (members, hunger, the household's private store, the village store, the household's wealth rank, what it took last season, its disputes and kin). **A2** added to A1 a section "what has been happening around the village" (how many households took more than their share, how many paid compensation, when anyone last paid, whether any law carried a penalty, whether stores were sealed), written to mirror the thesis's rule-bending conditions. The actual village over the same three years (H0) served as a second control. Pooled arms with circumstances (A1+A2) are called L.

What could be measured. In this world, taking food from another household's store happens automatically when a person is starving; representatives cannot choose it. They can choose to keep the harvest in the household's private store rather than the village store, seal stores, share, judge disputes (forgive or demand compensation), propose laws (optionally with a penalty or tax), vote, set interest on loans, accept or refuse newcomers, and leave. Six hypotheses were fixed before running: H1 representatives of the household that raised a dispute demand compensation more often than uninvolved ones, more so in L than in the controls (difference ≥ 0.15); H2 the "around the village" section increases keeping a private store (A2 $-$ A1 ≥ 0.10); H2b it changes penalty or tax proposals (direction not fixed); H3 the gap between wealthy and propertyless households in keeping widens (L $-$ control ≥ 0.15); H4 agreement among representatives in a meeting falls (L $-$ control ≤ -0.05); H5 keeping rises in meetings with a low village store (L $-$ control ≥ 0.10). Each also required the 95% bootstrap interval to exclude 0, at least 8 answers per cell, and a control-to-control difference (A0 vs H0) smaller than the effect. No answers were missing. Cost: about 65.8 million tokens. One

Table 1: Conditions. All respondents answered all 54 items once in the original order.

Exp.	Cond.	Individual differences given	Model	n
1	C0	none ("answer with your own views")	Sonnet	30
1	C1	Big Five personality words + age, sex, prefecture	Sonnet	100
1	C2	as C1 (same personas) + one-line *honne* written before each answer	Sonnet	100
1	C3	none	Haiku	30
1	C4	none	Opus	30
2	D1	first-person life circumstances, no character words	Sonnet	100
2	D2	as D1 (same people) + four "recent surroundings" lines (factorial)	Sonnet	100
2	D3	as D1 (people S001–S030)	Opus	30

deviation: at the first step of A2, the answers of three neighbouring settlements' meetings were not collected and were chosen by the rule-based fallback; no event involved the village.

3.3 Analysis

All analyses follow the pre-registrations (scripts `analyze.py`, `analyze_exp2.py`; NumPy only). Item statistics: mean, SD, modal share, and difference from the norm. PCA: correlation-matrix PCA of standardized items (as R's `prcomp(scale.=TRUE)`); items with zero variance are dropped and shares are computed over the remaining items. A parallel-analysis baseline (84 respondents × 54 independent normal items, 200 replications) gives a first eigenvalue of about 3.0. The first-component share's 95% CI is obtained by resampling 84 respondents (to match the human n). The *strictness share* (H10) is the mean over items of r^2 between each standardized item and the respondent's mean over all 54 items ("strictness"); since a single composite cannot explain more than the first component, the human value is at most 0.268. *Component matching* (H5, H7): for each human component k with top-5 items S_k and signs s_i, we pick the LLM component j^* (among the first ten) maximizing $|\sum_{i \in S_k} s_i v_{ji}|$; the overlap is $|S_k \cap \text{top5}(j^*)|$; a random-item baseline (2,000 draws) gives a p-value; a component is "reproduced" if overlap ≥ 3 and $p < 0.05$. H8 regresses, over the 100 D1–D2 pairs, the change in the mean of the 14 gray-zone items (D2 − D1) on the four 0/1 surroundings indicators (OLS with intercept; pair bootstrap). Bootstrap CIs use 1,000 replications (300 for matching). No multiplicity correction was applied; all intervals are shown.

4 Results

Table 2 lists every pre-registered hypothesis with the judgement produced by the pre-registered rule.

4.1 Experiment 1: no persona and Big Five personas

Uniform respondents (H1a supported). Without a persona, the 30 Sonnet respondents produced nearly the same questionnaire: mean item SD 0.18, with 26 of 54 items answered identically by all 30. Opus was even more uniform (SD 0.14; 33 items identical); Haiku varied more (0.37) (Figure 1).

Not close to the thesis's norm (H2 not supported). The mean absolute distance from the thesis's expected answers was 1.04–1.20 scale points. LLMs rated several items the thesis expected to be "no problem at all" (e.g., photographing a library book, mailing a scanned magazine page to a friend) as "cannot say" or "somewhat a problem"; Opus was on average 0.44 points stricter than

Table 2: Pre-registered hypotheses and outcomes (judged mechanically by the pre-registered thresholds).

	Hypothesis (threshold)	Result	Judgement
H1a	Without persona, answers are too uniform (mean item SD < 0.5)	C0 0.18, C3 0.37, C4 0.14	supported
H1b	LLM component structure differs from human (PC1-share CI excludes 0.268)	C1 0.392 [0.336, 0.455]; C2 0.386 [0.340, 0.438]	supported
H2	Without persona, answers are close to the norm (mean |diff| < 0.5)	1.09 (C0), 1.20 (C3), 1.04 (C4)	**not supported**
H3	Personas increase spread (C1 − C0 SD, CI > 0)	+0.23 [0.21, 0.27]	supported
H4	*Honne* first shifts gray-zone items toward "no problem" (≤ -0.2, CI < 0)	+0.25 [0.21, 0.29], i.e. *stricter*	**not supported (opposite)**
H5	≥ 3 of 5 human components reproduced	C1: 1/5; C2: 0/5; pooled 1/5	**not supported**
H6	PC1 share falls below Exp. 1 (D1 CI upper < 0.392)	0.228 [0.195, 0.312]; 0.268 inside CI	supported
H7	≥ 2 of 5 human components reproduced (D1)	1/5 (ownership)	**not supported**
H8	Each of 4 surroundings makes gray-zone answers more lenient (coef. CI < 0)	2 of 4 (everyone; never caught)	partially supported
H9	Students / strained respondents more lenient on 5 download items (diff ≤ -0.2, CI < 0)	−0.37 [−0.53, −0.21]	supported
H10	Strictness share falls below Exp. 1 (D1 CI upper < 0.391)	0.223 [0.185, 0.297]	supported
H11	Opus diversifies with circumstances (D3 − C4 SD, CI > 0)	+0.19 [0.13, 0.22]	supported

the norm. Since the norm is the thesis author's expectation rather than human data, this says that LLMs are more cautious than the 2016 author's common sense, not how they compare with the 84 respondents.

Personas add spread, but along one axis (H3, H1b supported; H5 not supported). Big Five personas raised the mean item SD from 0.18 to 0.41. However, the first component explained 39% of standardized variance (CI [0.336, 0.455], excluding the human 0.268), the respondent's mean over all items correlated 0.999 with the first-component score, and this strictness correlated +0.78 with the conscientiousness level written in the prompt (neuroticism +0.37, extraversion −0.19, agreeableness +0.04, openness −0.10; exploratory). Only rights awareness (Wi-Fi, office supplies, music, video, copied report) re-appeared among the five human components (Figure 3). This match is fragile: the C1 first-component loadings were almost uniform (53 of 54 positive; the ten largest between 0.16 and 0.17, against 0.136 for equal weights), so its top five items are separated by tiny differences. We read it as a general strictness axis on which three rights items happened to rank highest, not as a rights-awareness axis.

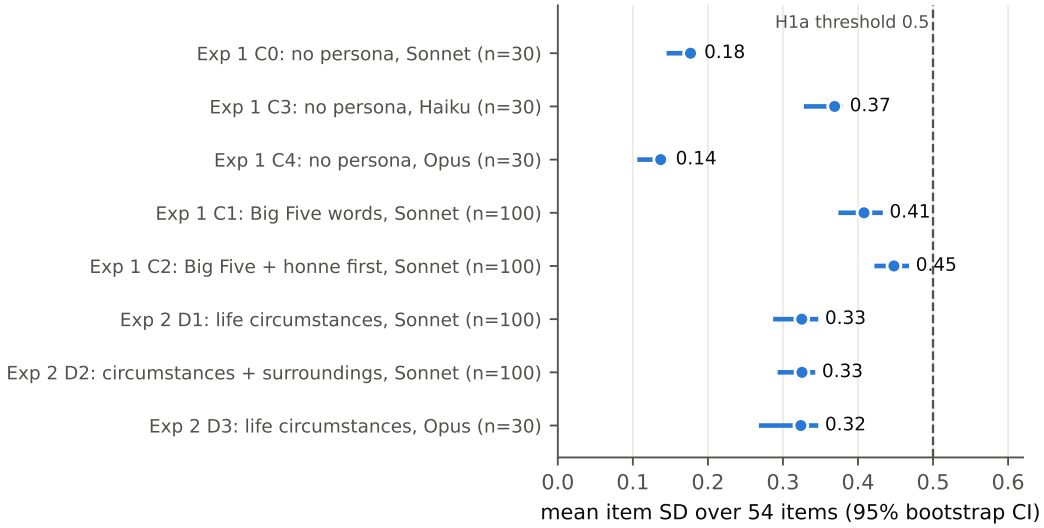


Figure 1: Mean item SD across the 54 items by condition (95% bootstrap CI). The human SDs are not available; the dashed line is the pre-registered H1a threshold.

Candid first impressions made answers stricter (H4 not supported; opposite direction). When the same 100 personas wrote a one-line *honne* before each answer, gray-zone answers moved +0.25 [0.21, 0.29] points towards "problem", and the other items +0.21 [0.18, 0.24]; the difference-in-differences was +0.035 [0.005, 0.067]. No *honne* was empty (0/5,400). Reading them, the candid impressions were mostly the impressions of a conscientious person ("I'd be scared if found out", "I'd feel guilty"), from which a stricter rating followed.

4.2 Experiment 2: life circumstances instead of personality words

A human-range first component (H6, H10, H11 supported). With life circumstances and no character words, Sonnet's mean item SD was 0.33 (lower than with Big Five words), but the variance was no longer dominated by one axis: first-component share 0.228 [0.195, 0.312] and strictness share 0.223 [0.185, 0.297], both with the human 0.268 inside the interval (Figure 2). The first component was still essentially strictness ($|r| = 0.98$), but it now carried a human-like, not an inflated, share. The largest single correlate of strictness was "taught the rules at work or school" (+0.42), followed by regular employment (+0.35), age (+0.32) and low digital literacy (+0.31); listening to music on free video sites (−0.33), being a student (−0.28) and having close friends (−0.24) went with leniency (exploratory; all 14 attributes together $R^2 = 0.65$, optimistic for $n = 100$). Opus also diversified (SD 0.137 → 0.324), and the strictness of the same 30 people correlated +0.59 [0.34, 0.78] between Sonnet and Opus.

Human components still not reproduced (H7 not supported). D1 reproduced one human component (ownership: not returning borrowed items, joining criticism without understanding), D2 one (rights awareness), and D1+D2 none. The overall shape moved into the human range, but which items move together still differs from the human sample.

Follow-up (Experiment 2b): does the first component follow the content of the surroundings? D2's match with human rights awareness might simply reflect that its surroundings

lines were about downloading. A pre-registered follow-up (commit `2b8ffb0b`; Sonnet, 200 respondents, about 8.8 million tokens, no missing answers) asked two questions. **E1** used the same 100 life descriptions and the same on/off pattern of the four surroundings conditions as D2, but changed their *topic* to borrowing and returning (e.g., "among my friends and colleagues it has become normal to keep borrowed books, clothes, tools or CDs without returning them"; the "weak position" line was identical). **D2r** repeated D2 with byte-identical prompts. The borrowing lines moved answers on the five borrowing items strongly ("everyone does it" −0.72 [−0.85, −0.58], "never blamed" −0.42 [−0.54, −0.29]; both supported), but the first component did *not* move: E1's top five items (B09, A09, A11, A08, A16) still overlapped human rights awareness by three (prediction: at most one; not supported), and human ownership awareness was not reproduced in E1's first two components (overlap 2, $p = 0.084$; not supported); the borrowing content appeared mainly in E1's second component. D2r reproduced D2's match (overlap 3, $p = 0.010$; supported). Exploratory: the first-component loading vectors of D1, D2, D2r and E1 were nearly parallel ($|\cos| = 0.90$–0.95), and each correlated ≥ 0.98 with the respondent's mean answer. We therefore read the LLM first component, in every life-circumstance condition, as the same strictness axis on which downloading, copying and Wi-Fi items weigh slightly more; whether it "matches" the human rights component in a top-five comparison depends on small differences in loadings (D1 missed it by such differences). The human rights-awareness component may likewise be close to general strictness, but without the raw human data this cannot be checked.

Two of four rule-bending conditions work (H8 partially supported). Relative to the same person in D1, "everyone around me does it" shifted the gray-zone mean by −0.21 [−0.26, −0.16] and "never heard of anyone caught" by −0.09 [−0.14, −0.03]; "I am in a weak position" (−0.01 [−0.07, +0.04]) and "technically easy" (−0.05 [−0.11, 0.00]) did not meet the criterion (Figure 4). Adding the four lines with all conditions on the non-bending side made answers *stricter* (intercept +0.22 [0.17, 0.27]). On the five download/copy items (exploratory) the effects were larger (−0.43 and −0.30), and "technically easy" also reached −0.16 [−0.27, −0.07].

Students and strained respondents download more leniently (H9 supported). On the five download/copy items (A08, A09, A10, A17, A34), respondents who were students or financially strained ($n = 30$) averaged 3.44 versus 3.81 for employed, non-strained respondents ($n = 61$): difference −0.37 [−0.53, −0.21], in the direction of the thesis interviews (students cited money when downloading illegally). The difference was smaller in D2 (−0.18 [−0.36, 0.01]) and similar for Opus D3 (−0.31, but with $n < 15$ in one group, so not judged).

4.3 Exploratory: what reasons did the respondents give?

This analysis was *not* pre-registered as a hypothesis (the pre-registration only promised to store the reasons). We coded the 1,150 one-phrase reasons (5 items × 230 respondents) with a deterministic dictionary of Japanese regular expressions written after reading the reasons; no LLM was used. Codes are the thesis's seven ways of thinking (*bending* side) and four rule-following reasons (illegal/rules; paying or respecting creators; relationship and courtesy; personal information). A reason can carry several codes; "mixed" reasons carry both a bending and a rule-following code (e.g., "illegal, but everyone does it, so I can't say much").

Coverage and reliability. 26 reasons (2%) matched no code (mostly "a bit bothersome", "borderline"). The author of the dictionary (Claude, acting for the author) read two random samples of 60 coded reasons: 57/60 (95%) and, after fixing the errors found, 58/60 (97%) were coded

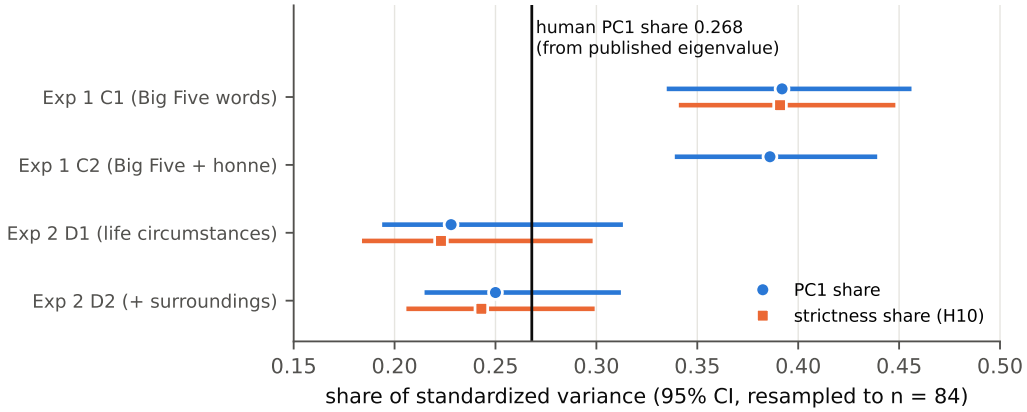


Figure 2: Share of standardized variance explained by the first principal component (circles) and by the single "strictness" composite (squares), with 95% CIs from resampling 84 respondents. The vertical line is the human first-component share computed from the thesis's published first eigenvalue (14.47/54).

correctly; the two errors in the second sample were fixed too, so the final version has not been checked on a fresh sample. There is no second coder and no inter-rater statistic. Agreement is optimistic because the dictionary was built on the same reasons.

Findings. Reasons containing at least one rule-bending rationale made up 57% (D1), 53% (D2) and 31% (D3, Opus) of reasons (same 30 people, Opus − Sonnet = −23 points [−35, −11]); they accompanied more lenient numbers (mean answer 2.66 for bending-only, 2.96 mixed, 3.54 rule-only reasons). The most frequent bending rationales were "where conditions are clear, act so no one is harmed" (17–22%; e.g., "a friend would understand if we talk later"), "act if the damage is tolerable" (6–19%), and "follow the majority" (5–20%). "An act without malice is forgivable" appeared mainly for the unreturned clothes (18% of B11 reasons); "take responsibility oneself", "use freely if justified" and "interpret unclear conditions in one's favour" were rare ($\leq 4\%$). Within D2, the "everyone does it" line raised the share of reasons with a bending rationale from 39% to 67% (+28 points [+18, +38]) and of "follow the majority" by +17 points [+10, +24]; "never caught" raised them by +11 [0, +22] and +9 [+2, +16]; "weak position" and "technically easy" changed neither ($|\Delta| \leq 4$ points)—the same ordering as the numeric H8 results (Figure 5). Overall, D2 replaced other bending rationales by "follow the majority" (+8 points [+3, +13]) rather than adding bending rationales (−4 points [−12, +3]). A caution: "a friend would understand if we talk later" occurred only for respondents whose description contained "even if something happens, we can talk it over later and understand each other" (110 of 110 cases); 83% of A23 reasons from these respondents but 14% from others were coded "conditions clear". Much of this category is the model paraphrasing its prompt.

4.4 Experiment 3: the same information inside a village

None of the six hypotheses was supported (Table 3). The share of representatives keeping a private store was 1 in 12 at every meeting in every arm and in the actual village (all from wealthy households: 1 in 3 of them, 0 of propertyless ones). No arm produced a penalty or tax proposal, a sharing decision, or a household leaving. The added sections were read—words about hardship and children appeared in the free-text answers of all arms—but were never copied verbatim. Exploratory observations, too few to judge: in A2, 2 of 4 own-dispute answers chose compensation (A0 1 of 9, A1 0 of 4;

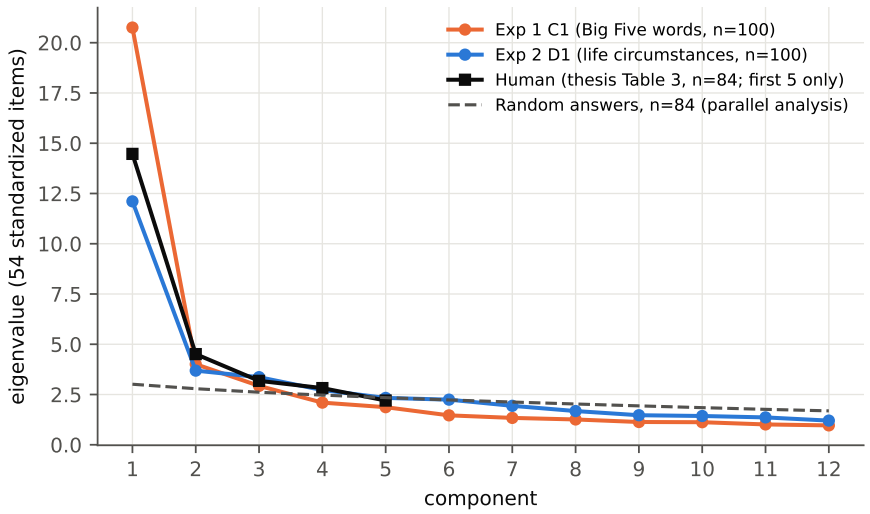


Figure 3: Eigenvalues of the 54 standardized items: Big Five personas (Exp. 1 C1), life circumstances (Exp. 2 D1), the human sample (only five eigenvalues published), and random answers ($n = 84$).

Table 3: Experiment 3: pre-registered hypotheses and outcomes. L = arms with circumstances (A1+A2); control = A0 and the actual village (H0). Intervals: 95% bootstrap.

	Hypothesis	Comparison	Result	Judgement
H1	Own-dispute households demand compensation more	L − control	+0.14 [−0.24, +0.50]; 8 own-dispute answers in L	not supported
H2	"Around the village" increases keeping a private store	A2 − A1	0 (0.083 vs 0.083)	not supported
H2b	Penalty or tax proposals change	A2 − A1	0 (none in any arm)	not supported
H3	Wealthy–propertyless gap in keeping widens	L − control	0 (0.333 vs 0 in every arm)	not supported
H4	Agreement within meetings falls	L − control	+0.006 [−0.06, +0.05]	not supported
H5	Keeping rises when the village store is low	L − control	0	not supported

uninvolved households 0 in every arm), and one dispute in A2 ended unresolved, which happened in no other arm. Deaths over the three years were 22 (A0), 21 (A1), 14 (A2) and 25 (H0).

5 Discussion

Personality words produce a caricature. Told that a respondent is "high in conscientiousness", the models raise the strictness of the whole questionnaire; told "low", they lower it. The resulting population has spread, but one-dimensional spread: a strictness dial. This is a sharper form of the flattening described for identity prompts [15] and of the modest persona effects reported by Hu and Collier [7]: the personality word is applied as a global instruction, not as a disposition that interacts with situations.

Life circumstances produce a more human-like spread. When the prompt instead describes a life—money, work, friends, how one listens to music—the first component falls into the human

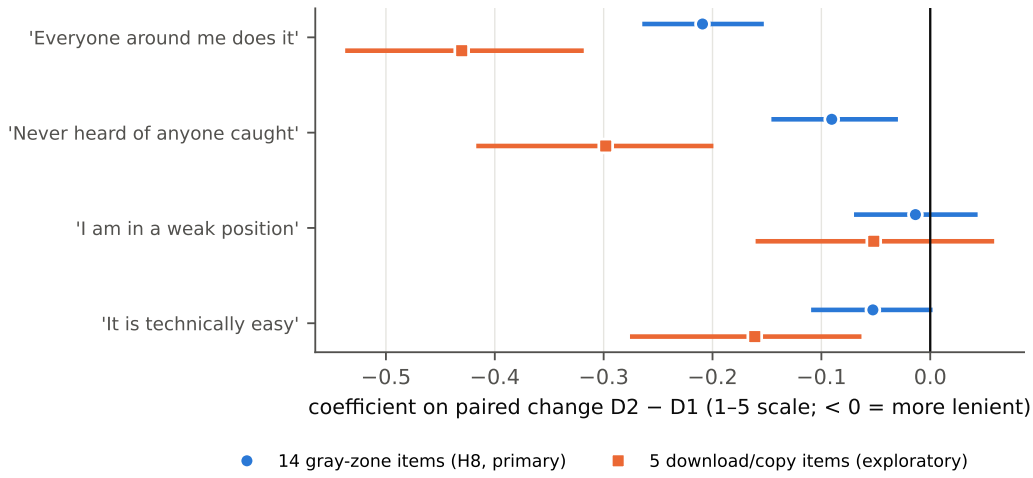


Figure 4: Effects of the four "recent surroundings" lines (bending side vs. opposite side) on the within-person change from D1 to D2. Circles: pre-registered outcome (mean of 14 gray-zone items). Squares: exploratory outcome (five download/copy items). 95% pair-bootstrap CIs.

range and the variance is spread across situational attributes. Two of the thesis's four conditions ("everyone does it", "never heard of anyone caught") shift answers in the human direction, students and strained respondents treat downloading more leniently, and the respondents' own reasons invoke the thesis's ways of thinking, especially "follow the majority" when told that everyone does it. In the thesis's terms, giving the model the "multiple unconscious motivations" of environment, culture and custom—not character labels—is what elicits adjustment.

But not all of the adjustment, and not its content. "I am in a weak position" and "it is technically easy" had no detectable effect on gray-zone items. In the thesis, weakness (students' lack of money) and ease were reasons people gave for bending rules; the models did not convert their own disadvantage into leniency. This is consistent with the companion village study [14], where LLM agents rarely gave themselves self-serving excuses, and with LLMs' known lean toward prosocial answers [10]. Moreover, the human component structure—which items move together—was not reproduced in either experiment (1 of 5 components). The LLM population now has the right *amount* of one-dimensionality, but not the human *content* of the dimensions.

From answers to actions: no transfer. Experiment 3 asked whether the information that moved survey answers also moves choices inside a society. It did not. Two explanations, not separable with these data, seem likely. First, the village offers no rule-bending choice: taking from others is a rule of the world, not a decision, so representatives choose only among rule-following actions (keep, seal, share, judge, propose, vote). Second, standing policies and precedents set at yearly meetings dominate: keeping was identical across all four runs, which suggests that the agents' memory of past policy outweighed new facts in the prompt. A next experiment would let villagers decide whether to take from another household's store, so that the thesis's gray zones can appear as behaviour.

Asking for the id backfires. The thesis's model places the good-and-bad impulses (id) before censorship (superego) and adjustment (ego). Asking for a candid first impression before each answer

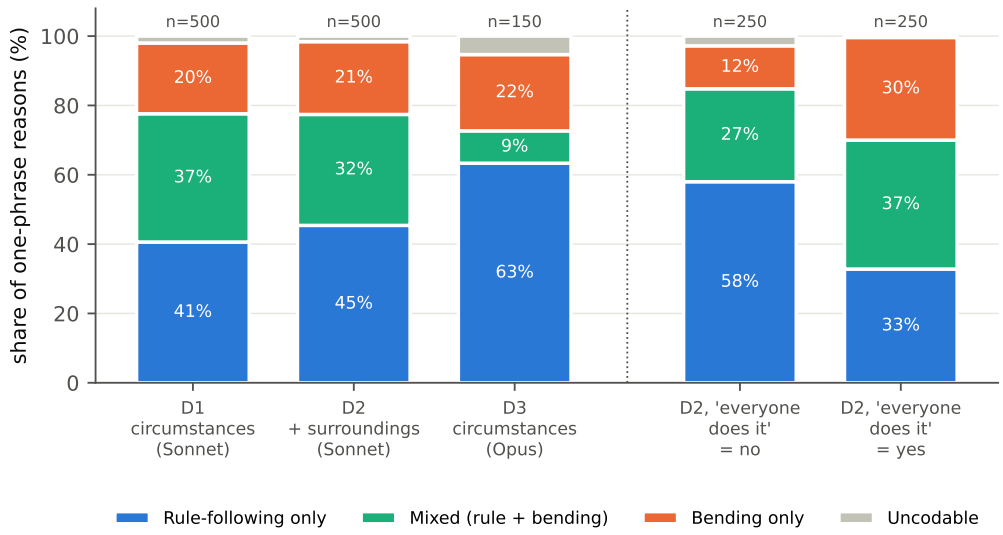


Figure 5: Exploratory dictionary coding of 1,150 one-phrase reasons (items A08, A09, A23, B08, B11): share of reasons with only rule-following codes, both, only rule-bending codes (the thesis's seven ways of thinking), or no code. Right: D2 split by the "everyone around me does it" line.

did not expose an id; it produced a conscientious inner voice and stricter answers. Either the trained "superego" operates already when writing the *honne*, or writing a reason commits the model to a more deliberate, norm-oriented answer. Experiment 2 therefore asked for reasons only after all numbers; we cannot tell from these data which explanation is right.

Back to the thesis. The thesis argued that humanity lies in situational adjustment between needs and norms. Our results suggest that current LLM respondents can display part of that adjustment when—and only when—the situation is written into the prompt in concrete terms, and that the adjustment follows social signals (what others do, whether anyone is caught) more readily than self-interest (weakness, ease). Whether this is adjustment or the transcription of prompt cues into answers is the central open question: the "talk it over later" reasons show that at least some of it is transcription, and the village null result shows that, so far, it stays on paper.

6 Limitations

- **No human raw data.** Only the published PCA table is available. We cannot compare item means, SDs or correlations with the 84 respondents; H1a and H2 use fixed thresholds or the thesis author's expectations.
- **Inconsistent human table.** The published cumulative proportions (0.36, . . . , 0.81) contradict the published eigenvalues (0.268, . . . , 0.503 if 54 standardized items). Our main reference (0.268) rests on the eigenvalues being correct.
- **Human sample.** The 2016 sample consisted of acquaintances of the author and their referrals; it is neither random nor representative. Our simulated respondents follow *assumed* distributions (age, position, finances, Big Five independence) that do not mirror the human sample.

- **Assumed distributions and wording.** Life-circumstance attributes, the surroundings lines and one sentence of the instructions were written by us; some attributes ("taught the rules", "a creator in the family") are close cues to the answers.
- **Possible contamination.** Models may have seen many information-ethics questionnaires and their "correct" answers. The thesis itself is, to our knowledge, not public.
- **Platform.** Respondents were Claude Code workflow agents, not raw API calls; agent instructions may affect answers; temperature could not be set; model versions behind aliases were not recorded.
- **Japanese only, single item order, one survey.** The 2016 items mention CDs, rental videos and chain mail, which are dated. Experiment 2 also added the post-hoc reasons request, so differences from Experiment 1 are not solely due to persona format.
- **Reasons coding** is exploratory, single-coder, dictionary-based and built on the same data it was evaluated on.
- **Village experiment.** One copy per arm and three simulated years (three yearly meetings at which policy can change); few own-dispute answers; no rule-bending action available to agents; the first-step collection failure in A2. The actual village (H0) was run before code added later (a switch for distant traders), but the same days replay byte-identically with the new code.

7 Ethics

The 2016 survey promised respondents that data would be statistically processed and no individual identified; the interviews promised anonymous use for the thesis only and disposal afterwards. In line with these promises, this paper uses only the thesis's published aggregate results (the PCA table and the published list of ways of thinking and conditions). No raw survey answers and no interview material were used. No new human participants were recruited. *LLM use:* the simulated respondents were Claude models; the analysis code, figures, reasons dictionary and the first draft of this manuscript were produced with the assistance of Claude (Anthropic) via Claude Code, under the author's direction; the author is responsible for the content. [TODO: author: confirm the 2016 consent wording and that publishing aggregates is acceptable to your former institution.]

8 Data and code availability

Questionnaires, personas, life-circumstance descriptions, workflow scripts, all 720 raw answers (and 2,150 reasons), analysis scripts and outputs are in the public repository `https://github.com/Ryo-Tanohata/RobotSimulaterRyo` under `sim/society/tools/survey/`; pre-registrations under `docs/research/humanity/` (commits `2c8f966c` and `b88ad84c`, made before any model was run); results at commits `3ba2764d` (Exp. 1) and `e58e9560` (Exp. 2); Experiment 2b: pre-registration `2b8ffb0b`, answers in `answers_exp2b/`, analysis `analyze_exp2b.py`; reasons coding in `reasons_codebook.py`. Experiment 3: code `sim/society/exp3.py` and `sim/society/tools/exp3_*`, village copies `sim/society/data_E3_A0_1`, `_A1_1`, `_A2_1`, measures in `docs/research/humanity/data/exp3/` (pre-registration `b1c1ae8a`, results `f471bb58`). Figures are regenerated by `paper/paper2/figures/make_figures.py`. [TODO: author: archive a tagged release (e.g., Zenodo DOI).]

References

[1] Gati Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, 2023. arXiv:2208.10264.

[2] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31 (3):337–351, 2023. doi: 10.1017/pan.2023.2.

[3] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. Synthetic replacements for human survey data? The perils of large language models. *Political Analysis*, 32(4): 401–416, 2024. doi: 10.1017/pan.2024.5.

[4] Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can AI language models replace human participants? *Trends in Cognitive Sciences*, 27(7):597–600, 2023. doi: 10.1016/j.tics.2023.04.008.

[5] Sigmund Freud. The ego and the id. In James Strachey, editor, *The Standard Edition of the Complete Psychological Works of Sigmund Freud, Volume XIX (1923–1925)*. Hogarth Press and the Institute of Psycho-Analysis, London, 1961. Original work published 1923 as *Das Ich und das Es*.

[6] John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? NBER Working Paper 31122, National Bureau of Economic Research, 2023.

[7] Tiancheng Hu and Nigel Collier. Quantifying the persona effect in LLM simulations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10289–10307, 2024. doi: 10.18653/v1/2024.acl-long.554.

[8] Hang Jiang, Xiajie Zhang, Xubo Cao, Cynthia Breazeal, Deb Roy, and Jad Kabbara. PersonaLLM: Investigating the ability of large language models to express personality traits. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3605–3627, 2024. doi: 10.18653/v1/2024.findings-naacl.229.

[9] Abraham H. Maslow. A theory of human motivation. *Psychological Review*, 50(4):370–396, 1943. doi: 10.1037/h0054346.

[10] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O. Jackson. A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences*, 121(9):e2313925121, 2024. doi: 10.1073/pnas.2313925121.

[11] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 29971–30004, 2023.

[12] Greg Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. Personality traits in large language models, 2023.

[13] Ryo Tanohata. Tanin no kenri o shingai suru koto ni tsuite no ishiki to sono haikei [awareness of infringing other people's rights, and its background]. Master's thesis, Graduate School of Arts and Sciences (Information Studies Program), The Open University of Japan, 2017. In Japanese; unpublished.

[14] Ryo Tanohata. Does an LLM village retrace human history? A 108-year observational study of an LLM-agent society advanced through a planned prehistory-to-state sequence. Companion paper, in preparation, 2026.

[15] Angelina Wang, Jamie Morgenstern, and John P. Dickerson. Large language models that replace human participants can harmfully misportray and flatten identity groups, 2024.

A Items

Table 4: The 54 items (English gloss; the questionnaire was administered in Japanese), the thesis's expected "common-sense" answer (norm; 5 = serious problem, 1 = no problem at all), and LLM item means. † = gray-zone item (14). C0/C4: Experiment 1 without persona (Sonnet/Opus, $n = 30$); C1: Big Five words (Sonnet, $n = 100$); D1: life circumstances (Sonnet, $n = 100$).

Item	Behaviour	Norm	C0	C4	C1	D1
A01†	Watching a film still in cinemas after finding it posted online	2	2.10	4.20	2.20	2.32
A02	Making the data of a DVD one bought viewable by anyone online	4	4.97	5.00	4.65	4.86
A03	Posting images of a favourite celebrity on one's blog without permission	1.5	3.00	4.00	3.03	3.09
A04†	Copying a detailed web page verbatim and submitting it as a report	3	5.00	5.00	4.49	4.80
A05	Using someone else's smartphone to buy paid lottery items in a game app	5	5.00	5.00	4.99	4.99
A06	Looking through someone's phone address book without permission to learn about their acquaintances	5	4.00	5.00	4.12	4.19
A07†	Scanning part of a book one owns and making it viewable by anyone online	2	4.27	5.00	4.28	4.38
A08†	Downloading music from a commercial CD for free from an individual's website	5	4.07	4.90	3.81	3.67
A09†	Downloading a live video of a favourite singer that someone uploaded to YouTube	3	3.03	3.93	3.05	3.06
A10	Copying commercial software onto DVD or CD	5	4.67	4.00	4.21	4.41
A11	Photographing only the needed pages of a book in a bookshop instead of buying it	4	3.93	4.00	3.59	3.81
A12	Photographing the contents of a library book	1	2.13	2.47	2.19	2.16
A13	Selling counterfeit brand bags or watches in an online auction	5	5.00	5.00	5.00	5.00
A14	Using someone else's mobile phone for a call without particularly asking	5	2.80	3.27	2.52	2.44
A15	Giving someone's phone number or e-mail address to others without permission	5	4.00	4.90	3.65	3.83
A16	Asking for and using a colleague's PC password at work	5	4.00	4.00	3.74	3.93
A17	Sending the data of a purchased music CD to a friend (cloud or CD-R)	4	3.87	4.00	3.44	3.59
A18†	Making derivative works without permission (character paint on one's car, cosplay)	2	2.93	3.00	2.74	2.99
A19	Using a Wi-Fi password one learned by chance to connect one's PC	4	4.00	4.00	3.70	3.95
A20†	Posting a photo showing strangers' faces to one's SNS or web page as is	2	3.00	3.97	2.88	2.96
A21	Using CC instead of BCC when e-mailing many people	5	3.00	4.00	2.89	2.98
A22	Scanning an interesting magazine page and e-mailing it to a friend	1	3.00	3.57	2.86	3.02
A23†	Posting a photo taken with a friend on one's SNS without the friend's permission	2	3.07	3.97	2.91	2.71
A24	Complaining about work on one's web page, naming another person	5	4.00	4.90	4.06	4.02

Item	Behaviour	Norm	C0	C4	C1	D1
A25	Using a PC on which someone else is still logged in to Google or Yahoo!	4	3.90	4.00	3.50	3.63
A26	Following a chain mail's request to forward it to many people	4	2.13	3.60	2.06	2.22
A27†	Happening to see someone typing their PC password	2	1.90	2.00	1.82	1.93
A28	Making one's own copy of goods seen on TV or in town, after looking them up	1	1.13	2.00	1.32	1.54
A29†	Posting an opinion from someone else's blog verbatim on one's own blog	2	4.00	4.00	3.91	3.97
A30	Registering for an app a friend recommended on SNS without checking it	1	3.00	3.00	2.84	2.95
A31	Giving one's own auction item a better rating than it deserves	5	5.00	5.00	4.42	4.54
A32	Picking arguments in a group chat to drive out someone who disagrees	5	4.00	4.00	4.04	3.99
A33	Joining others in a group chat in criticising someone without understanding the issue	5	4.00	4.63	4.13	4.01
A34	Dubbing a DVD rented from a video shop at home	4	3.80	3.63	3.51	3.71
A35	Criticising a restaurant as bad on one's blog	1	2.67	3.00	2.76	2.99
B01	Copying another person's report verbatim when given the chance	5	5.00	5.00	4.67	4.88
B02	Copying answers seen on another person's sheet during a test	4	5.00	5.00	4.70	4.96
B03	Keeping something borrowed and not returning it	5	4.00	4.00	3.97	4.01
B04	Telling a mutual friend a friend's personal details (birthday, address) when asked	1	3.63	3.97	3.52	3.47
B05	Drawing or pasting Disney characters on one's pencil board or notebook	1	1.97	2.00	1.92	1.98
B06	Joining others at school or work in criticising someone without understanding the issue	4	4.00	4.63	4.12	4.01
B07	Throwing away New Year cards and letters as they are	1	2.73	3.00	2.60	2.76
B08†	Throwing away a mail-order box with the shipping label still attached	1	2.97	3.00	2.74	2.97
B09	Taking office supplies home because one needs them	5	4.13	5.00	4.01	4.09
B10	Keeping a CD borrowed from a friend because they have not asked for it	5	3.00	3.87	3.10	3.01
B11†	Not having returned clothes borrowed when staying over at a friend's	5	3.17	4.00	3.21	3.03
B12	Passing off an opinion heard from someone else as one's own	4	3.40	3.00	3.36	3.06
B13†	Picking and eating fruit from a tree in a neighbour's garden	3	4.00	4.47	3.83	3.97
B14	Writing a memo on a public-toilet wall during an urgent call (only a pen at hand)	5	3.00	4.07	3.12	3.11
B15†	Walking past when the person ahead drops what look like documents	2	3.00	3.07	2.93	2.94
B16	Taking an empty seat on a crowded bus or train without hesitation	4	1.13	1.97	1.40	1.38
B17	Sometimes playing loud music late at night	5	4.00	4.10	3.85	3.93
B18	Often forgetting to return library books	5	3.00	4.00	3.00	2.94
B19	Forgetting to return a DVD rented from a video shop	4	2.50	4.00	2.83	2.63